

Analysis and experimental investigation of powder concentration and stirring velocity on powder mixed electrical discharge machining performance with 3D printed electrodes

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ABSTRACT

In this work, CuO mixed lemon peel dielectric is used along with the 3D printed aluminium electrodes to enhance the electrical discharge machining (EDM) performance. This work mainly focuses on improving the surface quality by optimizing the powder concentration and stirring velocity. A full factorial of 27 experiments is carried out with input parameters such as pulse off time with 3 levels (28, 58 and 98 μ s), powder concentration with 3 levels (1, 2 and 3 g/L) and stirring velocity with 3 levels (900, 1800 and 2700 rpm), while keeping the current, pulse on and voltage as constant. For all trials, the output responses viz. surface roughness (SR) and material removal rate (MRR) are noted. It is evidenced that EDM performance is influenced by the stability of powder mixed dielectric. To understand the most influential input parameter and its level, analysis of variance (ANOVA) and grey relational analysis (GRA) are carried out. The output analysis exhibited that powder concentration of 2 g/L, stirring velocity of 1800 rpm and pulse off time of 28 μ s as the best combination for the best quality of finish (4.72 μ m) with high MRR (0.0035 g/min) than the 1 g/L PC with 900 rpm SV for the best quality finish (5.87 μ m) along high MRR (0.0029 g/min) and 3 g/L PC with 2700 rpm SV for the best quality finish (5.38 μ m) with high MRR (0.0030 g/min) having pulse off time of 28 μ s. The experimental findings are validated with a 3D surface plot and SEM images taken on the machined surface. Also, a simulation model is built to understand the flow velocity profiles. This helps ensure the minimum stirring velocity to be maintained to enhance the stability of powder mixed dielectric.

1. Introduction

Electrical discharge machining (EDM) has given improved performance for applications requiring materials having complex shapes and extreme hardness, including high strength [1]. The occurrence of elevated temperature during the process has the potential to induce various defects on the workpiece surface [2]. A few examples of these flaws include pores, cracks, residual stress, and a recast or rebuilt layer that is incorrect. These defects are the result of intense heating and subsequent cooling that happens during the process [3]. The integration of powder additives in conventional EDM has led to the development of a novel machining technique known as powder mixed electric discharge machining (PMEDM). The primary objective of this approach is to enhance the efficiency and versatility of EDM by introducing small particles of powder into the dielectric solution, which is employed throughout the entirety of the machining process [4]. An inherent

benefit of PMEDM is its capacity to process materials that are too difficult to process by conventional EDM. Incorporating conductive particles enhances material removal and mitigates surface roughness during the machining of hard or composite materials [5]. This characteristic enhances the versatility of EDM, making it appropriate for a wider array of sectors such as aerospace, medical and automotive [6]. Enhanced surface finish and preciseness are also ensured by PMEDM. The conductive powders mixed in the dielectric play a crucial role in managing spark distribution. This, in turn, leads to improved surface finish and minimized recast layer, which is otherwise common issues in conventional EDM [7].

In recognition of its exceptional qualities viz. superior corrosion resistance, lightness and biological compatibility, Ti-6Al-4V has become widely utilized across multiple industries [8]. Despite these benefits, the increased strength and low thermal conductivity of Ti-6Al-4V present significant obstacles to traditional machining

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methods such as excessive wear of the tool, inadequate smoothness of the surface or subpar surface finish [9]. In this scenario, EDM would be an ideal solution for making complex shapes with this material [10].

To make EDM machining more efficient, various researchers explored various techniques that assist in thermal-electrical properties. The addition of biodegradable oils, oil emulsion and conductive or non-conductive particles resulted in several changes to the dielectric. Besides, the tool electrode was also enhanced via production methods or thermal treatments such as quenching or cryogenics [11]. Due to the strong reliance on electrical science, many parametric methods were attempted to attain the desired objectives by altering the sparking phenomenon [12,13]. Since the late 1990s, titanium alloy machining using EDM has garnered scientific attention, mostly via experimental investigations. The most popular research subjects were MRR, TWR, and SQ and how machining circumstances like pulse-on current, pulse-on duration, dielectric fluid, and electrode material affect them [14]. Productivity in machining Ti-6Al-4V using EDM is directly dependent on the workpiece temperature during machining. Therefore, selecting the appropriate machining parameters is crucial. The energy distribution between the electrode and workpiece is a primary factor leading to the reduced material removal rate (MRR) and excessive tool wear rate (TWR) when machining titanium alloys using electrical discharge machining [15]. The study aims to evaluate the performance of a machined titanium alloy when subjected to a Copper electrode in the presence of neem and kerosene oils as dielectrics. Based on the findings of the study, it has been observed that dielectric neem oil exhibited a higher level of machining performance in comparison to EDM oil [16]. Investigating the machining responses that occur when normal water is combined with deionized water to explore the potential for environmentally friendly electrical discharge machining (EDM). The most important variables in the titanium alloy's sustainable EDM process, according to the scientists, are discharge current and duty factor [17]. Earlier researchers devised experiments using the central composition plan, which included discharge current, pulse length, duty cycle, and graphite powder concentration. Research revealed that discharge current was the most significant factor. Experiments confirmed graphite powder's pleasing impact of reducing surface roughness by 27 % [18]. Adding particles to EDM dielectric liquid reduces its break-down strength, boosting early spark and machinability. Powder-mixed electrical discharge machining (PMEDM) faces challenges in maintaining powder-dielectric mixture homogeneity and minimizing negative effects on pump impellers for a wide range of powder sizes. An attempt has been made to overcome obstacles and constraints in PMEDM by creating a novel mixing mechanism, and the machining trials were conducted to study its effect on the material removal rate (MRR) of Titanium alloy Grade 5 [19]. Murahari and Kumar [20] observed a significant reduction in surface roughness with the usage of a powder-mixed dielectric medium. The authors concluded that the machined surface was primarily affected by the concentration, conductivity and size of the powder. In another experimental work [21], the usage of silicon powder mixed dielectric medium helped in achieving reduced surface roughness, crater size and white layer thickness (WLT) than the conventional EDM. Chaudhari et al. [22] study aimed to examine the effects of varying concentrations of alumina (Al₂O₃) powder on the MRR, TWR and Surface Roughness (SR) while performing the EDM Machining process of Nitinol. The authors concluded alumina powder concentration, T_{off} and T_{on} as crucial inputs which considerably impacted all the output responses. The alumina addition helped in achieving enhanced MRR and reduced TWR and SR. A significant enhancement in surface morphology, as exhibited by SEM analysis was attributed to the presence of alumina nanoparticles. Kumar et al. [23] examined the machinability of carbon fibre-reinforced polymers while considering the impact of SiC powder. Through their research, the authors discovered that a SiC powder concentration of 4 g/L, a pulse current of 1 A, and a duration of 15 μ s yielded the lowest surface roughness, measuring at 1.2 μ m. Davis et al. [24] improved the surface characteristics of the biocompatible magnesium

alloy using Zinc (Zn) powders in PMEDM. A significant decrease in dimension deviation DD and an increase in MRR was reported for Zn powder mixing ratio of 4 g/L. The comparative experiments at a concentration of 4 g/L evidenced the improved responses with copper electrodes over the brass and significant improvements such as reduced resolidified debris, cracks and pores.

The literature research discloses that many endeavours have been made to meliorate EDM performance by the employment of diverse tool electrodes, parametric optimization, incorporation of different conductive/non-conductive powder mixed dielectrics and even bio dielectrics. To the best of the author's knowledge, there is limited literature that deals with sustainable PMEDM that embraces the combination of sustainably manufactured tool electrodes with green dielectric mixed with different nanopowders. Also, only a handful of literature is available that studies the combined impact of powder concentration and stirring velocity using bio-dielectrics. Additionally, there is no literature available for lemon peel dielectric, CuO powder and stirring velocity optimization effect on MRR and SR. The novelty of the present work lies in the optimization of stirring velocities and powder concentration for a lemon peel biodiesel based dielectric to obtain the highest degree of surface finish without a compromise on MRR. Thus, the present work adds value in the domain of process improvement, sustainable manufacturing and eco-friendly working standards. Thus, the aims of this study have been carefully formulated to address the gaps in existing literature. The objectives that have been constructed are as follows: (i) investigating the effect of lemon peel biodiesel dielectric mixed copper oxide (CuO) nanopowders (ii) determining the optimum powder concentration to achieve the high SR (iii) examining the combined effect of stirring velocity and powder concentration on SR, MRR and (iv) multi-objective optimization of input parameters for optimum surface finish and MRR.

2. Materials and methods

EDM is predominantly adopted in highly precise applications where top priority is given to intricate profiles with superior surface finish over material removal rate and/or tool wear rate. The intricate shapes can be easily imparted to AlSi10Mg electrodes with 3D printing techniques. Moreover, the AlSi10Mg electrodes are economical and lighter than most of the traditional electrodes. Due to these distinct advantages, the authors adopted the 3D printed AlSi10Mg electrodes over copper and graphite electrodes. In this study, aluminium alloy (AlSi10Mg) is considered the electrode material due to its promising electrical and thermal characteristics and manufactured through 3D printing. The aluminium electrode is produced by Selective Laser Melting (SLM) equipment (SLM 280, Treatstock, Germany). It is equipped with a 400 W laser source. The other specifications related to the printing are as follows: (i) laser power output: 470 W, (ii) scanning speed: 1300 mm/s, (iii) spacing between the hatches is 0.19 mm, (iv) controlled chamber environment: shielded with argon gas and (v) the layer thickness is 30 μ m. The 3D printed tool electrode used in the research has an outer diameter of 10 mm. The workpiece utilized in this study is a titanium grade 5 alloy. Table 1 and Table 2 illustrate the composition of the

Table 1
Chemical constituents of the titanium alloy grade 5.

Elements	Wt. (%)
Nb	0.04
Ni	0.001 max
Fe	0.112 max
C	0.369 max
Cu	0.02 max
V	4.22 max
Al	5.48 max
Ti	Balance

Table 2
Chemical constituents of the aluminium alloy electrode.

Elements	Wt. (%)
Si	9 to 11
Cu	0.05 max
Fe	0.55 max
Mn	0.45 max
Mg	0.2 to 0.45
Zn	0.10 max
Ni	0.05 max
Al	Balance

electrode and workpiece materials respectively. Table 3 presents the physicochemical properties of dielectrics derived from lemon peel biodiesel. Biodiesel is widely recognized as an environmentally friendly (green) dielectric due to its minimal environmental impact and the reduced harm to operators. CuO nanopowder is chosen to mix with lemon peel biodiesel dielectric as its excellent thermal conductivity would help to improve EDM performance. The purchased nanopowders (Sigma-Aldrich, Purity: 99 %) are determined to possess an average particle size of 25–35 nm. Fig. 1 displays the morphology and XRD analysis of the CuO powder particles adopted in this research.

Three distinct concentrations of powder-mixed dielectrics were created by combining the appropriate amount of CuO nanopowders with the lemon peel biodiesel dielectric. Fig. 2 depicts the various steps involved in the preparation of nanopowder mixed dielectrics. First, CuO nanopowder is poured into the lemon peel dielectric, which is subsequently subjected to magnetic-stirring for 10 min. This is accompanied by the sonication process for 30 min, which facilitates the disintegration of agglomerates or clusters of nanoparticles, thus ensuring their homogeneous dispersion in the fluid. The uniform distribution of particles is essential for attaining the desired characteristics such as increased heat transfer efficiency or better electrical conductivity and improved effectiveness from nanopowder mixed dielectrics [5]. At last, the prepared powder-mixed dielectric is taken out to conduct the machining experiments.

The stability of nanopowder mixed with green dielectric was checked by determining the duration during which the particles stay suspended throughout the dielectric medium without clumping together as shown in Fig. 3(a-d). The freshly prepared dielectric medium is exhibited in Fig. 3(a). After about 75 min, the nanopowder started to agglomerate at the top as shown in Fig. 3(b). The agglomeration intensified after 120 min as depicted in Fig. 3(c) and after 180 min powder started to settle at the bottom as shown in Fig. 3(d).

Table 4 displays the different levels selected for the input parameters. The detailed research methodology of the current work is presented in Fig. 4, whilst the schematic of the adopted die-sinking PMEDM setup is portrayed in Fig. 5. The tests are carried out using three different dielectrics with varying powder concentrations at different operating levels. Table 5 gives the full factorial design which gives every possible combination of input variables. The output responses viz. MRR and SR are measured after every machining experiment. Each experiment is performed thrice, and the average is calculated to serve as a typical value. To determine the MRR, the disparity in workpiece weight is

measured before as well as after the machining operation. This is accomplished via a digital weighing balance manufactured by (Mettler Toledo, United States). Eqn (1) is used to calculate the MRR in this research [25].

$$MRR = \left[\frac{(W_i - W_f)}{t} \right] \quad (1)$$

where MRR in gram/min, W_i – Weight of the workpiece at the beginning in grams, W_f – Weight of the workpiece after machining in grams; t – machining duration in min; ρ_w – Workpiece density in gram/mm³

The surface roughness is quantified using a Marsurf XR20 surface profilometer manufactured in Germany, while Keyence 3D Microscope (VHX 7000 N) is utilized to get the topography of the machined surfaces. The surface morphology images are obtained using a FEI QUANTA 250 FEG microscope (Thermo Fisher Scientific, US).

3. Results and discussion

The output responses such as MRR and SR obtained at each operating condition is presented in Table 6. Every experiment is conducted three times to get the average value for representation.

3.1. Multi-response optimization

The desirable outcomes of any machining process will be the maximum material removal rate with the minimum surface roughness. Based on the contradictory nature of the responses, it is necessary to employ a multi-response optimization technique to determine the optimum level for input parameters to attain a positive impact on output responses. Thus, the grey relational analysis, a multi-objective approach widely utilized in various fields due to its ability to achieve balanced compromise is adopted in the present work [4]. The normalization of experimental results (SR and MRR) is the initial step in grey relational analysis. The connection between the target and real experimental data is described by the grey relational coefficient, which is computed from the normalized experimental data. After that, the grey relational grade (GRG) is determined by taking the average of the grey relational coefficient for each process response. The grey relational grade evaluates the overall excellence of multiple process outputs. Thus, the simplification of optimizing several complex process responses may be achieved by focusing on optimizing a single grey relational grade. The strategy involves expressing the linear data preprocessing procedure for MRR, which is a higher-the-better metric, as an Eqn. (2).

$$x^*i(k) = \frac{x_i(k) - \max x_i(k)}{\max x_i(k) - \min x_i(k)} \quad (2)$$

Furthermore, the SR is optimal when it is minimized. This is mathematically represented by Eqn. (3)

$$x^*i(k) = \frac{\max [x_i(k)] - [x_i(k)]}{\max [x_i(k)] - \min [x_i(k)]} \quad (3)$$

Here $i = 1, 2, 3, 4 \dots m$ and $k = 1, 2, 3, 4 \dots n$; m is the total number of experiment data whereas n is the total number of responses. $x_i(k)$ represents the original sequencing, $x_i(k)$ represents the sequencing, $\max x_i(k)$ Indicates the greatest value of $x_i(k)$ and $\min x_i(k)$ Indicates the minimum value of $x_i(k)$, and x is the expected value.

The grey relational coefficient is computed using Eqn. (4)

$$\varepsilon_i(k) = \frac{\Delta_{\min} + \varepsilon \Delta_{\max}}{\Delta_{oi}(k) + \varepsilon \Delta_{\max}} \quad (4)$$

$\Delta_{oi} = ||x_o(k) - x_i(k)||$, the term $x_o(k)$ indicates the reference order whereas $x_i(k)$ refer to comparability order. $\Delta_{\min}$ and $\Delta_{\max}$ are the absolute minimum and maximum differences.

The grey relational grade is determined by combining the grey

Table 3
Lemon peel biodiesel dielectric properties.

Property	ASTM Standard	Lemon Peel Biodiesel
Density (g/ml)	ASTM D1298	0.853
Viscosity (cSt)	ASTM D445	2.54
Thermal conductivity (W/mK)	ASTM D 2717-95	0.176
Specific heat (kJ/kgK)	ASTM D2890-92	1.86
Breakdown voltage (kV)	ASTM D1816-12	29
Oxygen content (% wt.)	ASTM D3176	1.51
Flashpoint (°C)	ASTM D93	64

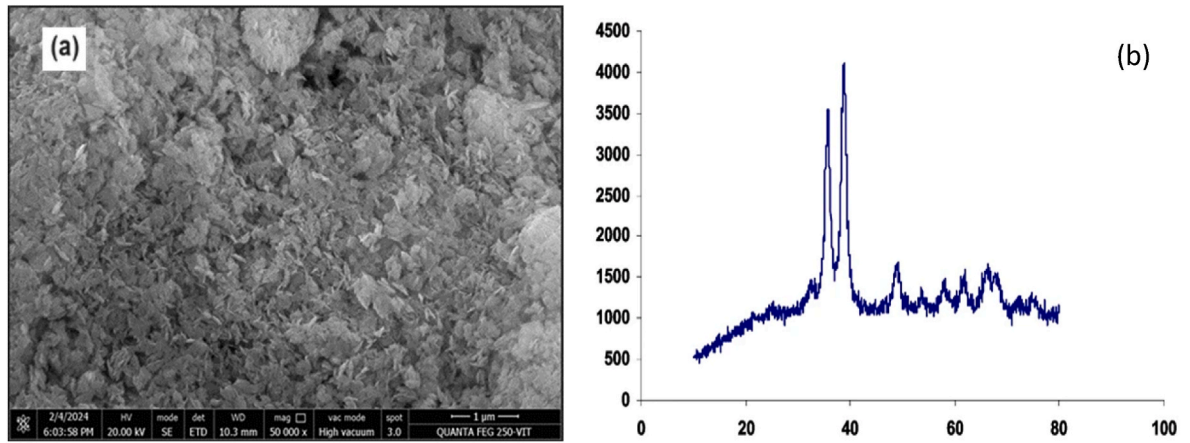


Fig. 1. SEM and XRD profile of CuO.

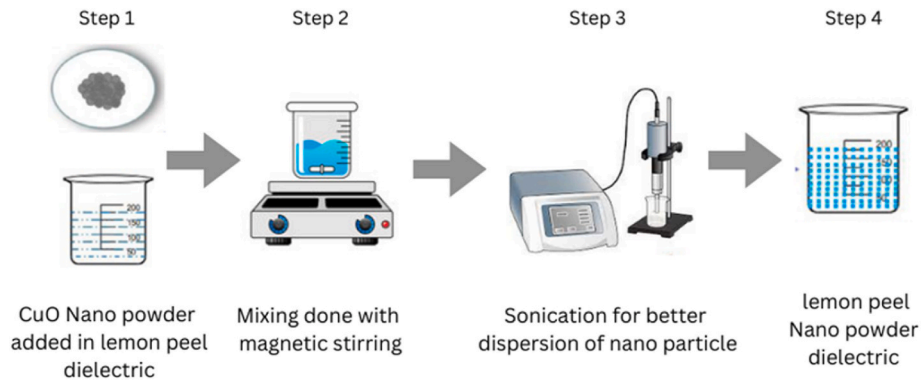
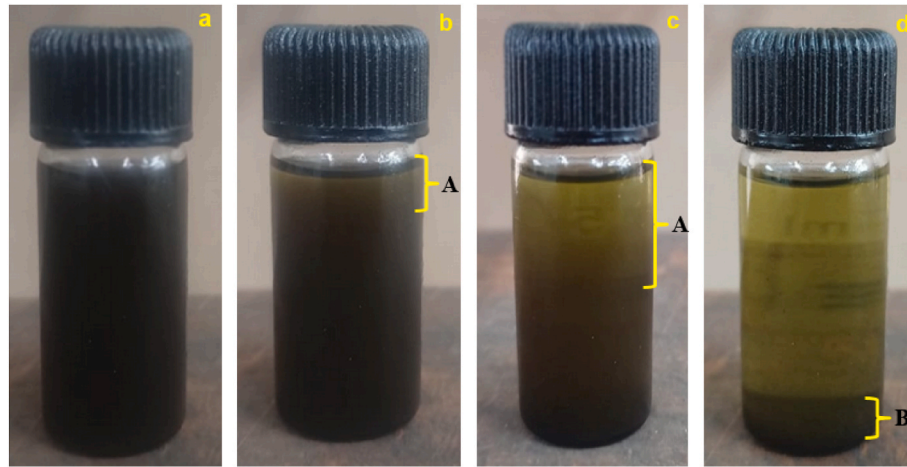


Fig. 2. Steps involved in nanopowder mixed dielectric preparation.



A – Agglomeration of particles ; B- Settling of particles

Fig. 3. Stability of CuO nanopowder mixed lemon peel biodiesel dielectric after (a) 0 min (b) 75 min (c) 120 min (d) 180 min.

relational coefficients as shown in Eqn. (5)

$$\gamma_i = \frac{1}{n} \sum_{k=1}^n \varepsilon_i(k) \quad (5)$$

where γ_i is the desired grey relational grade for the i^{th} iteration, n is the total number of response characteristics.

In this case, the ranking is determined by the highest values of the

grey relational grade. The optimal parametric settings are shown at the highest-ranked level. Table 7 presents a comprehensive overview of the experimental data, encompassing the normalized sequence, deviation sequence, grey relational coefficients (GRCs), grey relational degree (GRD), and ranking for each of the conditions that were subjected to testing. This multi-objective optimization concluded that a powder concentration of 2 g/L powder concentration is the best to obtain high MRR and low SR. The top three ranks are also obtained for the same

Table 4
Input parameters and their levels.

Parameters	Levels
Pulse-off time (μ s)	28,58,98
Pulse-on duration (μ s)	100
Duty factor (%)	78, 64, 51
Current (A)	20
Supply voltage (V)	60
Powder concentration (g/L)	1,2,3
Polarity	Positive
Stirring velocity (rpm)	900, 1800, 2700

powder concentration.

3.2. Analysis of variance of SR and MRR

Analysis of variance (ANOVA) is adopted in this research to

determine how different machining parameters affected the output characteristics. The equations are generated, and the ANOVA analysis is performed using Design Expert software. The significance of the variables is evaluated with an accuracy level of 95 % showing the efficacy of model terms. When the P -value is less than 0.05, it indicates that the machining variable has a significant impact on determining the responses [26].

The outcomes of ANOVA for the response SR are presented in Table 8. The Model F-value of 23.14 implies the model is significant. There is only a 0.01 % chance that an F-value this large could occur due to noise. The model terms with a P -value less than 0.05 indicated that they are significant. In this model, A, B, AC, A^2 and C^2 are significant terms and those with values greater than 0.1 are not significant. This analysis is in line with the results reported by previous researchers [1]. The highest F-value (64.44) is registered for AC and concluded that the combination of powder concentration and stirring velocity is the most influencing factor. This is then followed by B, pulse off time (F-value =

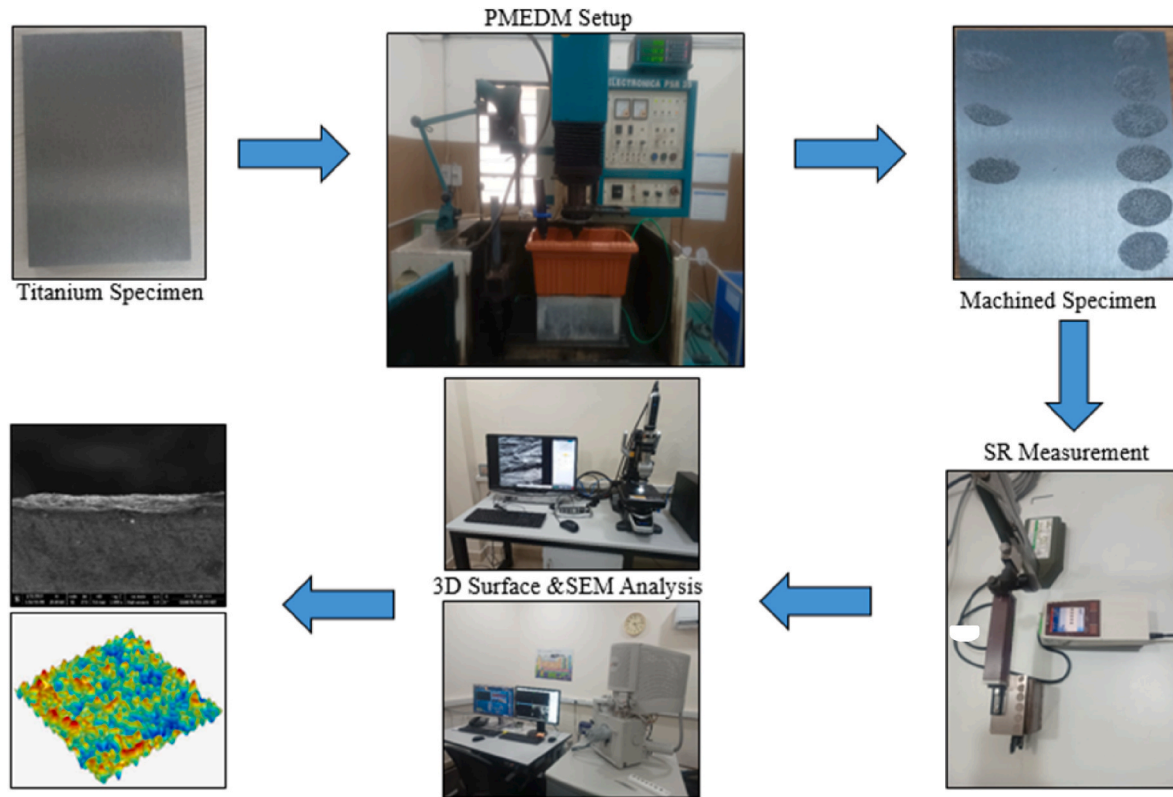


Fig. 4. Graphical illustration of experimental methodology.

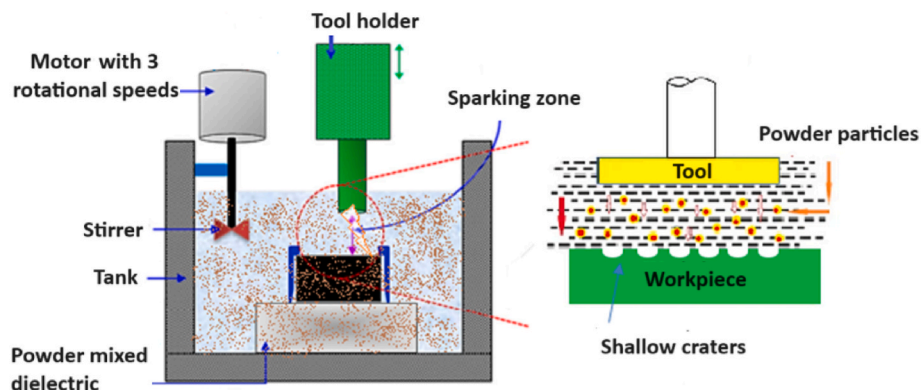


Fig. 5. A schematic representation of the die-sinking PMEDM adopted in this work.

Table 5

Full factorial design.

Exp. No	Powder concentration (g/L)	Pulse-off time (μs)	Stirring velocity (rpm)
1	1	28	900
2	1	28	1800
3	1	28	2700
4	1	58	900
5	1	58	1800
6	1	58	2700
7	1	98	900
8	1	98	1800
9	1	98	2700
10	2	28	900
11	2	28	1800
12	2	28	2700
13	2	58	900
14	2	58	1800
15	2	58	2700
16	2	98	900
17	2	98	1800
18	2	98	2700
19	3	28	900
20	3	28	1800
21	3	28	2700
22	3	58	900
23	3	58	1800
24	3	58	2700
25	3	98	900
26	3	98	1800
27	3	98	2700

Table 6

Calculated MRR and measured SR under different machining conditions.

Exp. No	Powder concentration (g/L)	Pulse-off time (μs)	Stirring velocity (rpm)	SR (μm)	MRR (g/min)
1	1	28	900	5.87	0.0029
2	1	28	1800	6.03	0.0025
3	1	28	2700	6.65	0.0021
4	1	58	900	5.29	0.0026
5	1	58	1800	5.71	0.0022
6	1	58	2700	6.37	0.0019
7	1	98	900	4.81	0.0024
8	1	98	1800	5.21	0.0021
9	1	98	2700	6.01	0.0018
10	2	28	900	5.75	0.0025
11	2	28	1800	4.72	0.0035
12	2	28	2700	5.48	0.003
13	2	58	900	5.41	0.002
14	2	58	1800	4.52	0.0028
15	2	58	2700	4.98	0.0024
16	2	98	900	5.03	0.0015
17	2	98	1800	4.21	0.0024
18	2	98	2700	4.9	0.0021
19	3	28	900	6.49	0.0022
20	3	28	1800	5.89	0.0027
21	3	28	2700	5.38	0.0030
22	3	58	900	5.94	0.0016
23	3	58	1800	5.33	0.002
24	3	58	2700	5.13	0.0026
25	3	98	900	5.61	0.0007
26	3	98	1800	5.02	0.0011
27	3	98	2700	4.85	0.0014

56.97) while considering SR as a desirable characteristic.

ANOVA analysis is extended towards MRR to gain more insight into the influencing factors. This is presented in Table 9. The Model F-value of 20.54 implies the model is significant. There is only a 0.01 % chance that an F-value this large could occur due to noise. In this model, the terms A, B, C, AB, AC, A² and C² are significant. From the analysis, it is observed that B is the most influencing factor (F-value = 90.06) which is succeeded by AC, the combination of powder concentration and stirring

Table 7

Outcomes of grey relational analysis.

Step 1		Step 2		Step 3		Grey Relational Grade	Rank
MRR	SR (μm)	MRR	SR (μm)	MRR	SR (μm)		
-0.214	0.480	0.214	0.520	0.700	0.490	0.595	10
-0.357	0.393	0.357	0.607	0.583	0.452	0.518	17
-0.500	0.000	0.500	1.000	0.500	0.333	0.417	26
-0.321	0.653	0.321	0.347	0.609	0.591	0.600	9
-0.464	0.483	0.464	0.517	0.519	0.492	0.505	21
-0.571	0.133	0.571	0.867	0.467	0.366	0.416	27
-0.393	0.820	0.393	0.180	0.560	0.735	0.648	5
-0.500	0.633	0.500	0.367	0.500	0.577	0.538	14
-0.607	0.400	0.607	0.600	0.452	0.455	0.453	23
-0.357	0.333	0.357	0.667	0.583	0.429	0.506	20
0.000	0.750	0.000	0.250	1.000	0.667	0.833	1
-0.179	0.543	0.179	0.457	0.737	0.523	0.630	6
-0.536	0.567	0.536	0.433	0.483	0.536	0.509	18
-0.250	0.930	0.250	0.070	0.667	0.877	0.772	3
-0.393	0.743	0.393	0.257	0.560	0.661	0.610	8
-0.714	0.727	0.714	0.273	0.412	0.647	0.529	15
-0.393	1.000	0.393	0.000	0.560	1.000	0.780	2
-0.500	0.770	0.500	0.230	0.500	0.685	0.592	11
-0.464	0.260	0.464	0.740	0.519	0.403	0.461	22
-0.286	0.400	0.286	0.600	0.636	0.455	0.545	13
-0.071	0.610	0.071	0.390	0.875	0.562	0.718	4
-0.679	0.423	0.679	0.577	0.424	0.464	0.444	24
-0.536	0.627	0.536	0.373	0.483	0.573	0.528	16
-0.321	0.693	0.321	0.307	0.609	0.620	0.614	7
-1.000	0.533	1.000	0.467	0.333	0.517	0.425	25
-0.857	0.730	0.857	0.270	0.368	0.649	0.509	19
-0.750	0.787	0.750	0.213	0.400	0.701	0.550	12

Table 8

Statistical analysis of SR.

Source	Sum of Squares	DF	Mean Square	F-value	P-value	Significant or not
Model	8.87	9	0.9858	23.14	<0.0001	significant
A-PC	0.2906	1	0.2906	6.82	0.0182	significant
B-Pulse off	2.43	1	2.43	56.97	<0.0001	significant
C-Stir velocity	0.0082	1	0.0082	0.19	0.6660	
AB	0.0060	1	0.0060	0.14	0.7128	
AC	2.75	1	2.75	64.44	<0.0001	significant
BC	0.0688	1	0.0688	1.61	0.2211	
A ²	2.49	1	2.49	58.38	<0.0001	significant
B ²	0.0411	1	0.0411	0.96	0.3400	
C ²	0.8239	1	0.8239	19.34	0.0004	significant
Residual	0.7244	17	0.0426			
Cor Total	9.60	26				
R ² = 92.45 %, R ² _{adj.} = 88.46 %, R ² _{pred.} = 83.27 %						

velocity (F-value = 38.32).

Generally, high R² values indicate that the model is both adequate and exhibiting the effectiveness of the developed model. R² = 0.9245 is observed in SR analysis with a predicted R² of 0.8327 is in reasonable agreement with the adjusted R² of 0.8846. In MRR analysis also, the observed R² = 0.9128 shows a reasonable level of agreement with the Predicted R² (~0.8327) and adjusted R² (~0.8846). The difference between the Predicted R² and Adjusted R² values is less than 0.2 observed in both SR and MRR analysis, which is in an acceptable range. The regression formulae for determining SR and MRR are presented in Eqns. (6) and (7)

$$\begin{aligned} \text{SR} = & 4.6784 - 0.127275 * A - 0.367222 * B - 0.0214077 * C + 0.0222297 \\ & * AB - 0.478333 * AC + 0.0754392 * BC + 0.643889 * A^2 + 0.0847454 * \\ & B^2 + 0.370556 * C^2 \end{aligned} \quad (6)$$

Table 9
Statistical analysis of MRR.

Source	Sum of Squares	DF	Mean Square	F-value	P-value	Significant or not
Model	9.652E-06	9	1.072E-06	20.54	<0.0001	significant
A-PC	5.579E-07	1	5.579E-07	10.69	0.0045	significant
B-Pulse off	4.702E-06	1	4.702E-06	90.06	<0.0001	significant
C-Stir velocity	2.663E-07	1	2.663E-07	5.10	0.0373	significant
AB	1.238E-06	1	1.238E-06	23.70	0.0001	significant
AC	2.001E-06	1	2.001E-06	38.32	<0.0001	significant
BC	7.508E-10	1	7.508E-10	0.01	0.9060	
A ²	7.350E-07	1	7.350E-07	14.08	0.0016	significant
B ²	3.177E-08	1	3.177E-08	0.61	0.4461	
C ²	2.400E-07	1	2.400E-07	4.60	0.0468	significant
Residual	8.876E-07	17	5.221E-08			
Cor Total	0.0000	26				
$R^2 = 91.28\%$, $R^2_{adj} = 87.12\%$, $R^2_{pred} = 80.44\%$						

$$\text{MRR} = 0.00255046 - 0.000191892 * A - 0.000494444 * B + 0.000106306 * C - 0.000296396 * AB + 0.000383333 * AC + 0.0000157 * BC - 0.000366667 * A^2 + 0.000055 * B^2 - 0.000216667 * C^2 \quad (7)$$

The residual plots provide a visual representation of the validation of the satisfactory results obtained from ANOVA [27]. ANOVA analysis is believed to be reliable and appropriate for the model if it meets certain assumptions. This is ascertained by looking into the residuals, which show the discrepancy between the expected and experimental responses [28]. Fig. 6(a) and Fig. 7(a) show a linear association between normal probability and externally studentized residuals; this suggests that for the best model performance, the data points should follow the normal distribution. In this case, a 45° angle would indicate that the data is linear [29]. The suitability of the models is assessed by looking at the residuals and the distribution plot as shown in Figs. 6(c) and Fig. 7(c). The evaluation shows how well the plotted points lined up with the hypothesized line of normality to determine the suitability and importance of our model [30].

Similarly, Figs. 6(b) and Fig. 7(b): display that there are no discernible patterns or trends in the experimental data and that the residuals were random concerning the expected response. Based on the scatter plot, it is evident that there is no consistent pattern in the data, which supports the unusual nature of the residuals [31]. Figs. 6(d) and Fig. 7(d) exhibit a dispersed, random distribution when the residuals are plotted against the run number. These findings are crucial in assessing the adequateness and plausibility of the used model [32].

3.3. Significance of pulse-off time

The impact of pulse-off time on MRR and SR can be better comprehended through the main effect plot. This is illustrated in Fig. 8. The responses exhibited a declining trend with an increase in pulse-off time. When the pulse-off time increased from 28 to 98 μs , keeping the powder concentration at 2 g/L and stirring velocity at 1800 rpm, the MRR was reduced by about 31 % while the surface roughness was improved by 10.8 %. Increased pulse-off time extends the duration between the consecutive sparks thus decreasing the number of active sparks and released energy. Reduced discharge energy causes a reduction in thermal energy. As a result, the work material is subjected to less melting and vaporization [33] contributing to reduced MRR. The result outcomes are in good agreement with previous studies [34]. The increased

pulse-off time was determined to produce a positive impact on SR. An increase in pulse-off time ensues in reduced discharge energy and thermal energy, which subsequently results in smaller crater formation. The increased time available between consecutive sparks might have improved the flushing efficacy [4,34] thus resulting in the overall smoothness of the machined work material. The results are aligned with the previous research work [36].

3.4. Significance of PC and stirring velocity combination

The ANOVA analysis revealed that the combination of powder concentration and stirring velocity has a significant impact on the MRR and SR. The inclusion of nanopowder in the dielectric enhances heat dissipation, resulting in enhanced machining performance and surface finish. Besides, the powder-mixed dielectric has the potential to remove debris from the machining zone, thus resulting in improved stability and precision in EDM operations [37]. The effect of powder concentration and stirring velocity on SR and MRR are shown in Fig. 9(a) and Fig. 9(b) respectively. The distribution of the plasma channel gets improved by several pathways such as (i) reduction in electrical resistivity of the dielectric medium through the powder particles and (ii) reduced impulse forces in the discharge gap due to the fragmentation of powder particles disrupting the spark channel. All these effects cumulatively lead to an even distribution of spark energy, which ensures enhanced MRR and reduced SR [35]. At increased powder concentration (~3 g/L or more) in dielectric, considerable reduction in MRR is evidenced with poor surface finish. This might be associated with process instability caused by arc circuiting. This ultimately affects the machining cycle, leading to surface deterioration and reduced MRR [38]. The process instability associated with PMEDM must be overcome using some external mechanism to refrain from adverse effects. Maintaining the stirring velocities in the dielectric tank above the critical range to get over the agglomeration is needed. The increased stirring speeds typically improve heat transfer by facilitating the distribution of nanopowders into the dielectric liquid, preventing sedimentation and enhancing convective heat transfer. Nevertheless, extremely high speeds can result in the clumping together of nanoparticles, which can diminish their efficacy [39]. The grey relational analysis concluded that 2 g/L is the optimum powder concentration to machine Ti-6Al-4V to facilitate high material removal with a better finish. The outcomes of these experimental findings can be validated by comparing them with the 3D surface plots generated on the EDM surfaces. The R_z value, a deterministic characteristic of the peak and valley height within the sample area is minimum in Fig. 10(b) as compared to Fig. 10(a) and (c). This is due to the optimum PC and stirring velocity combination which makes nanopowder mixed dielectric more stable contributing to an efficient machining cycle which improves the EDM efficiency with a smoother surface [39]. Also, the convective heat transmission within the powder mixed dielectric is often enhanced by optimizing the stirring speed. This is because optimum stirring velocity facilitates the mixing of fluids and introduces new fluids to the heat transfer surface, both of which improve the efficiency of heat exchange [40]. It is also observed that for the same condition when stirring velocity is low compared to optimum stirring velocity SR is increased to $\approx 22\%$ and MRR is reduced to $\approx 29\%$. This is due to a lack of heat transfer due to low stirring velocity and a less efficient flushing cycle. When the stirring velocity is above the optimum stirring velocity SR is increased to $\approx 16\%$ and MRR is reduced to $\approx 15\%$. This is due to high or excessive stirring which will lead to agglomeration of nanoparticles which will affect the effectiveness of the nanofluid reducing its performance.

3.5. Response Surface Methodology

Response Surface Methodology (RSM) is a powerful tool that helps in optimizing processes by analyzing the connections between various factors and the outcomes they produce. It utilizes statistical and

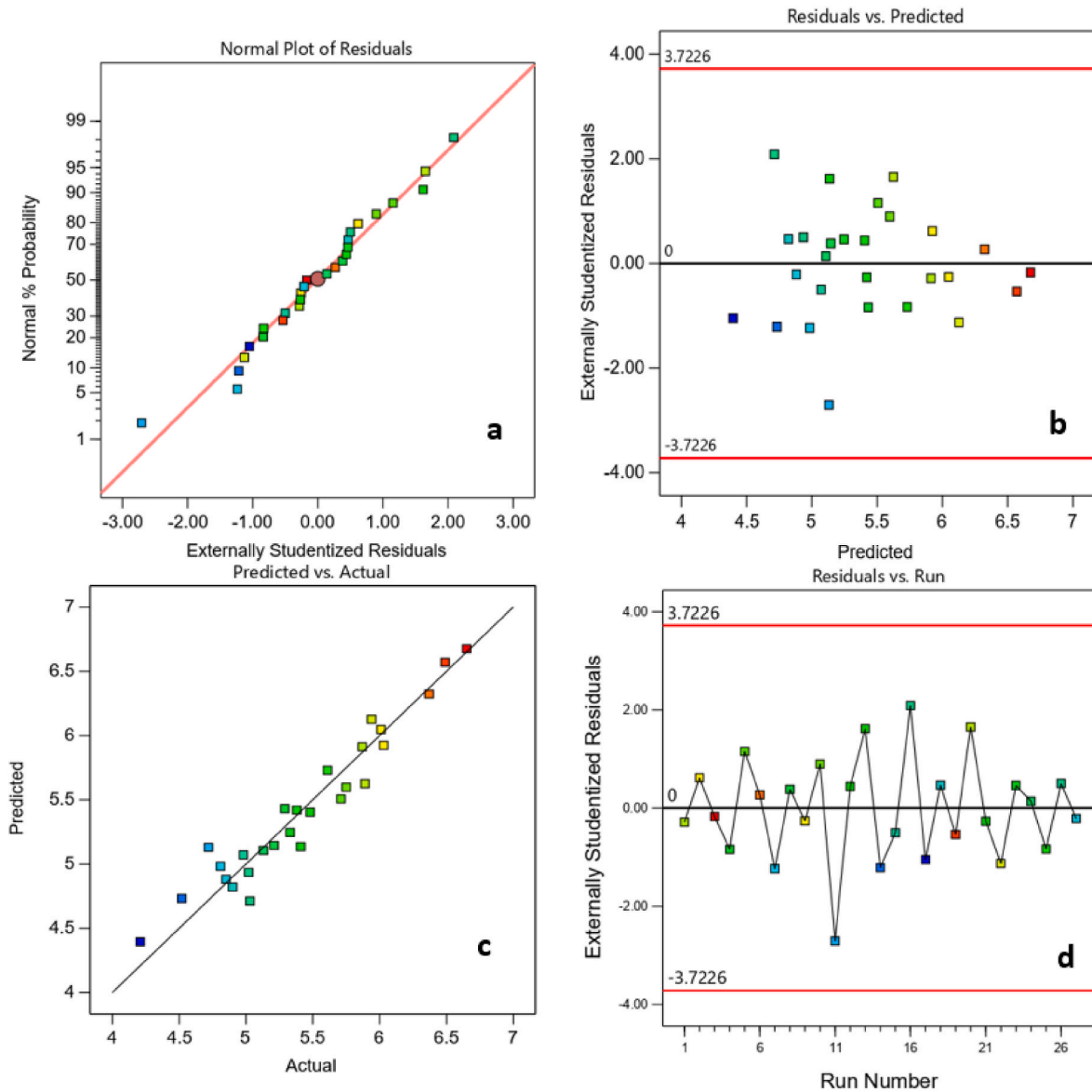


Fig. 6. (a–d). Residual plots for SR.

mathematical techniques to uncover these relationships and make informed decisions [41]. The main objective of RSM is to ascertain the optimal configuration of input variables to achieve the highest level of performance for the response variables. Additionally, RSM aims to construct an empirical model that accurately characterizes the system, assess and measure the interaction effects between different input variables, and develop a predictive model for forecasting responses under diverse conditions [42]. The significance of Response Surface Methodology (RSM) lies in its capacity to minimize the required number of experimental runs for investigating the impacts of numerous factors, therefore resulting in time and resource savings. It aids in optimizing processes to improve product quality and consistency, resulting in substantial cost savings in materials, labour, and time [43]. RSM offers a thorough examination of how different factors interact and impact the outcome, allowing for improved control of the process [44].

Response surface methodology (RSM) is a tool used to generate contour plots that relate the influential parameters and the output response [45]. In this research, the output responses considered are MRR and SR while the stirring velocity and powder concentration are considered the influential parameters. This has been attempted to evaluate the interdependence between the three parameters. i.e. stirring

velocity, powder concentration and either SR or MRR. SR was reduced from $5.87 \mu\text{m}$ to $4.72 \mu\text{m}$ when the stirring velocity increased from 900 rpm to 1800 rpm and powder concentration from 1 g/L to 2 g/L. During this change, MRR increased from $0.0029 \text{ mm}^3/\text{min}$ to $0.0035 \text{ mm}^3/\text{min}$. Further increase in powder concentration and stirring velocity was found to produce unfavourable results. When the stirring velocity was increased from 1800 rpm to 2700 rpm and powder concentration from 2 g/L to 3 g/L, SR increased from $4.72 \mu\text{m}$ to $5.38 \mu\text{m}$ and MRR reached $0.0030 \text{ mm}^3/\text{min}$ from $0.0035 \text{ mm}^3/\text{min}$. This is evidenced by Fig. 11 (a–d).

3.6. Simulation modeling

To achieve a homogeneous dispersion of powder particles, it is imperative to ensure that the liquid flow remains in a continuous upward motion starting from the bottom and evenly distributed. During the stationary condition of the working fluid, the powder particles exhibit settling movement in the liquid. The liquid Reynolds number $Re < 1$ is obtained from the Stokes equation, and the kind of motion velocity may be computed using the following Eqn. (8) [46].

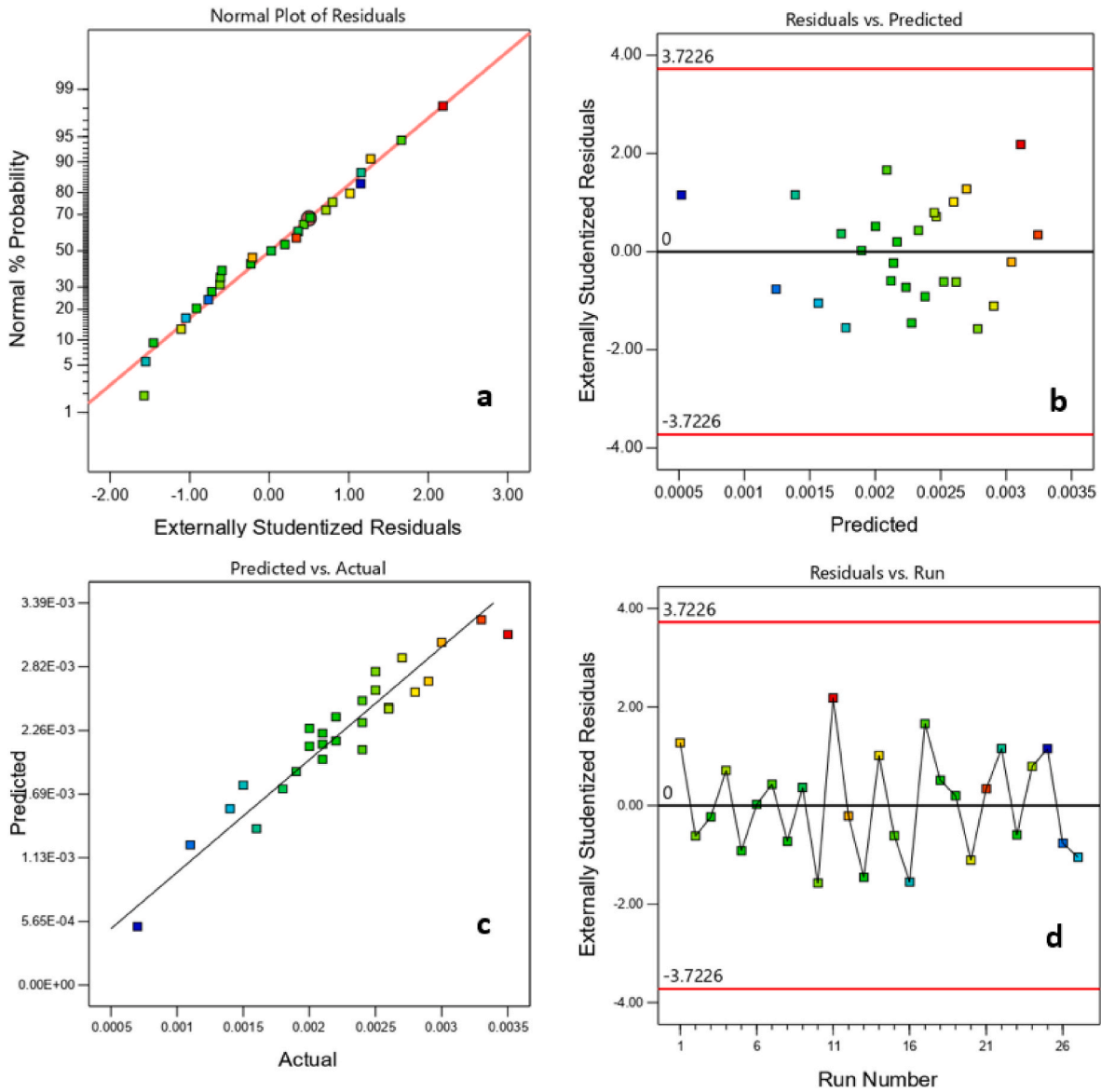


Fig. 7. (a–d). Residual plots for MRR.

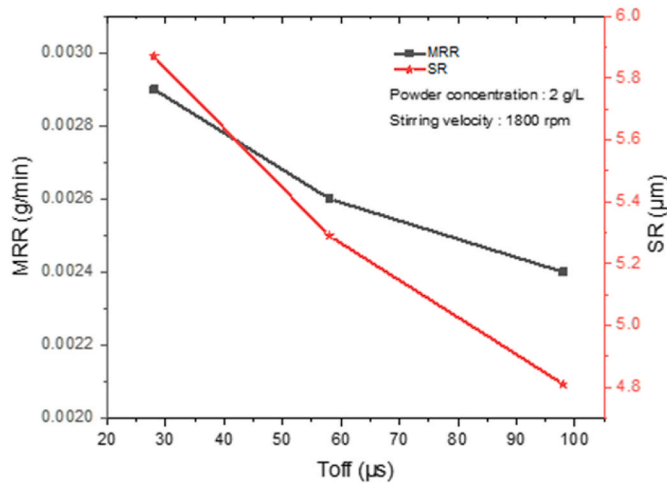


Fig. 8. Effect of pulse-off time on MRR and SR.

$$v_0 = \frac{2}{9}(\rho_1 - \rho_2)g\alpha^2 / \mu \quad (8)$$

Where v_0 – settling velocity in m/s, ρ_1 and ρ_2 – densities of powder particle and dielectric in kg/m^3 , μ – viscosity of dielectric in Pa.s, α – radius of nanoparticle in m, g – gravity in m/s^2 .

The settling velocity of a solid particle remains constant after its size and type have been established. The smaller radii particles settle more slowly and stay suspended more at a certain stirring speed. In theory, solid particles may be suspended if the liquid flow rate is equal to or higher than their settling velocity. With a density of 6315 kg/m^3 , the spherical CuO nanopowder diameter α was determined to be 25 nm. At a temperature of 20°C , the lemon peel's dielectric density is less than or equal to 853 kg/m^3 , with a kinematic viscosity of $1.06 \text{ mm}^2/\text{s}$ and a dynamic viscosity of 0.904 cP . By using the formula, the settling velocity of CuO nanopowder particles was calculated to be $8.230 \times 10^{-9} \text{ m/s}$. In this investigation, the concentration of solid particles is less than 5 %, and the settling velocity correction coefficient f_w is 0.84. The settling velocity formula is presented below: and the obtained velocity is $6.913 \times 10^{-9} \text{ m/s}$

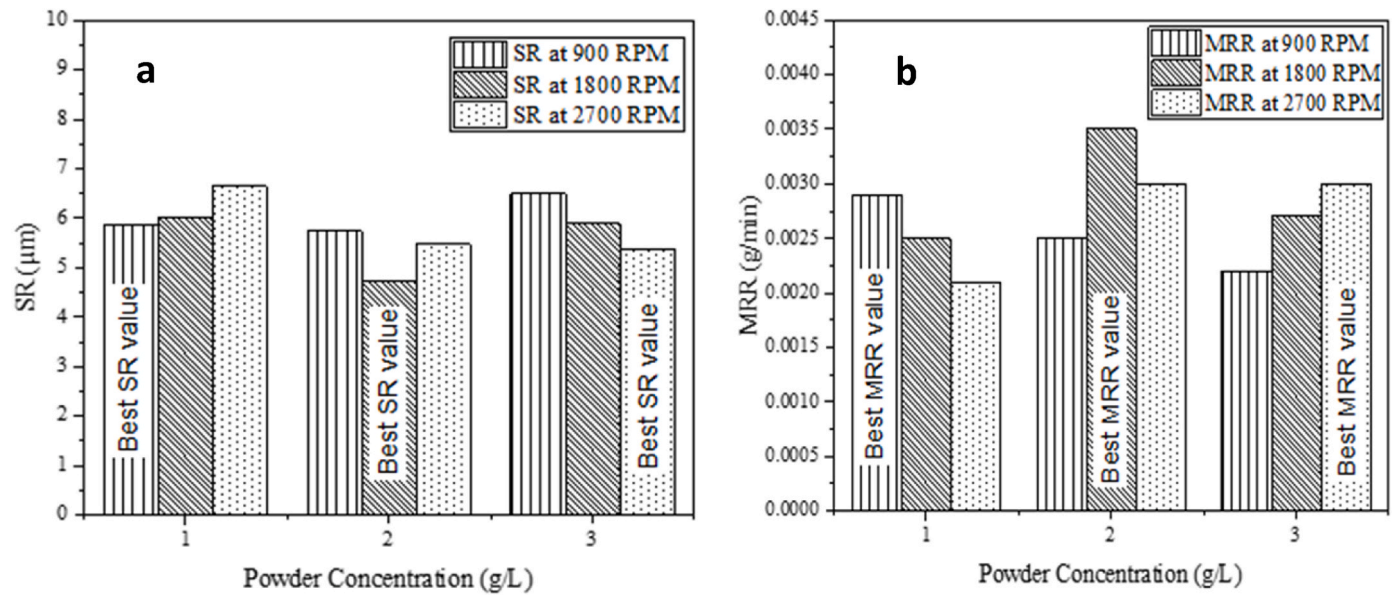


Fig. 9. Effect of powder concentration and stirring velocity on (a) SR and (b) MRR.

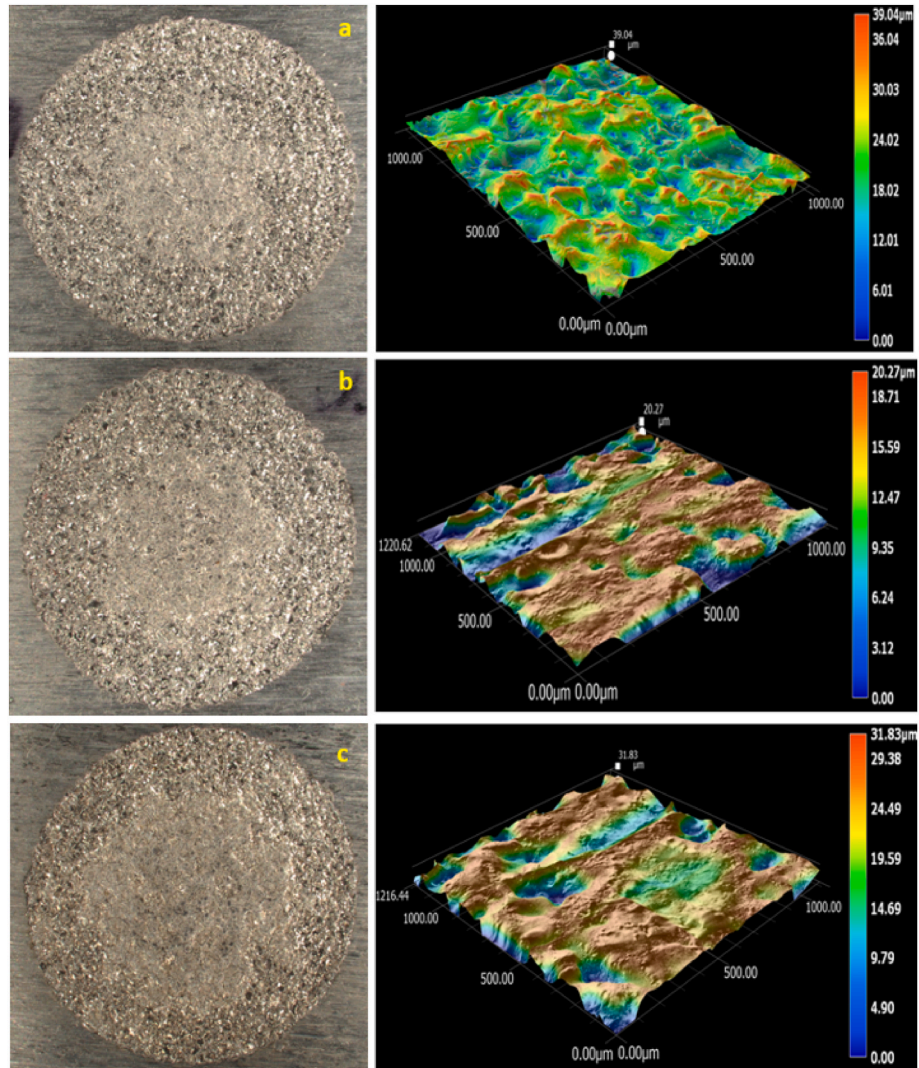


Fig. 10. (a–c). 3D surface plot analysis with different stirring velocities (a) 900 rpm (b) 1800 rpm and (c) 2700 rpm stirring velocity (Condition: PC - 2 g/L; Pulse off time: 28 μs)

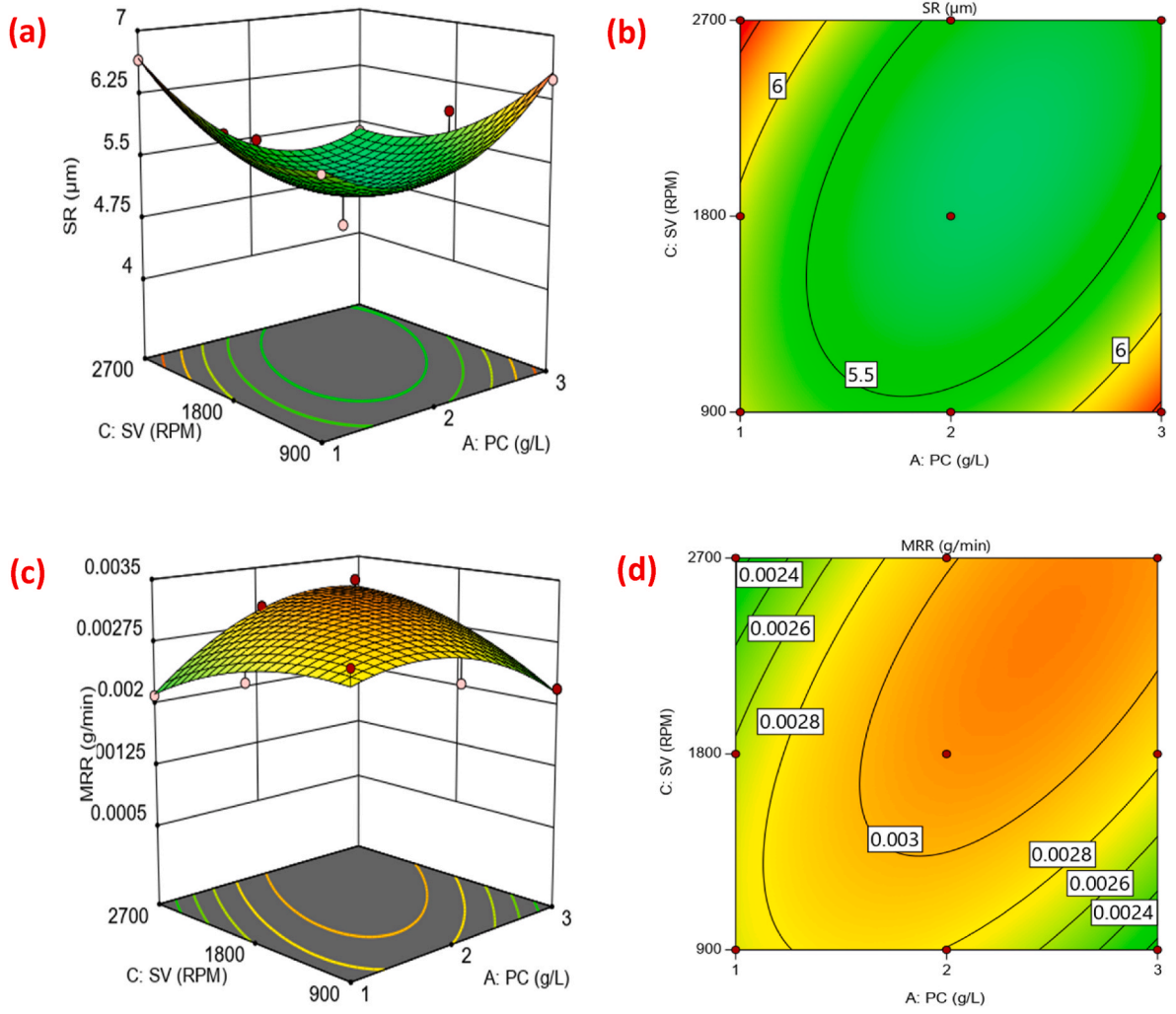


Fig. 11. (a,b) Surface and Counter plots for SR and (c,d) Surface and Counter plots for MRR.

$$v = f_w v_0 \quad (9)$$

To understand the fluid flow velocity inside the stirring tank for Different stirring speeds. The following considerations are made to conduct the simulation research investigation.

1. The walls of the mixing tank, including the base, are classified as stationary walls.
2. The uppermost surface of the mixing tank is designated as a free surface region and is classified as symmetrical. Considering stirring is done at the centre of the tank.
3. The wall and base of the mixing tank are defined as stationary walls, whereas the stirrer and stirring shaft are considered revolving walls.
4. Stirring speed is maintained uniformly during each stirring cycle.

While conducting a numerical simulation of liquid flow, the study focused on the impact of stirring velocity to obtain homogeneous dispersion, mixing, and suspension. Utilizing the Euler-Euler flow model. The following Eqns. 10–17 represent the basic concepts of solid and liquid 2-phase flow in this model [39].

For conserving volume fractions:

$$\alpha_l + \alpha_s = 1 \quad (10)$$

for the continuity of liquid in a phase:

$$\frac{\partial(\alpha_l \rho_l)}{\partial t} + \nabla \cdot (\alpha_l \cdot \rho_l \cdot \mu_l) = 0 \quad (11)$$

for the continuity of solid in a phase:

$$\frac{\partial(\alpha_s \rho_s)}{\partial t} + \nabla \cdot (\alpha_s \cdot \rho_s \cdot \mu_s) = 0 \quad (12)$$

The movement of liquid in a system:

$$\frac{\partial(\alpha_l \rho_l \mu_l)}{\partial t} + \nabla \cdot (\alpha_l (\rho_l \mu_l \mu_l)) = \alpha_l \nabla \rho + \nabla \cdot \mathcal{T}_l + \alpha_l \rho_l g + k_{sl}(\mu_s - \mu_l) \quad (13)$$

The movement of solids in a system

$$\frac{\partial(\alpha_s \rho_s \mu_s)}{\partial t} + \nabla \cdot (\alpha_s (\rho_s \mu_s \mu_s)) = \alpha_s \nabla \rho + \alpha_s \nabla \rho + \nabla \cdot \mathcal{T}_s + \alpha_s \rho_s g + k_{sl}(\mu_l - \mu_s) \quad (14)$$

Relationship between the momentum of a solid and a liquid

$$k_{sl} = \frac{3}{4} C_D \frac{\alpha_s \alpha_l \rho_l |\mu_s - \mu_l|}{d_s} \alpha_l - 2.65 \quad (15)$$

$$C_D = \frac{24}{\alpha_l Re_s} \left[1 + 0.15 (\alpha_l Re_s)^{0.687} \right] \quad (16)$$

The following equation gives the Reynolds number:

$$Re_{sl} = d_s \rho_l |\mu_s - \mu_l| / \mu_s \quad (17)$$

From Fig. 12 (a-d), it is observed that all three stirring speeds have achieved the minimum stirring requirement (i.e. v_0) which confirms that during stirring, the particle settlement to base was not happening. From the simulation results, it is observed that the flow pattern remains identical at all stirring speeds but any increase in stirring velocity results in increased flow velocity.

3.7. Surface morphology

The morphology of surfaces machined by PMEDM at different stirring velocities is investigated using scanning electron microscopy and presented in Fig. 13(a-c). The EDM process requires precise control of several parameters to provide a flawless machined surface which is devoid of any imperfections such as micro cracks, holes and globules. At the powder concentration of 2 g/L and pulse-off time of 28 μ s, the machined surface obtained with powder-mixed dielectric stirred at 1800 rpm is found to produce minimal surface flaws compared to dielectric stirred at 900 rpm and 2700 rpm. The surface is free from microcracks, craters, recast layers and pores. This is depicted in Fig. 13 (b). Improper energy distribution in the machined area may produce poor surface quality in terms of deeper and broader craters due to the concentration of heat [26]. This is demonstrated in Fig. 13(c). More optimized energy distribution and increased process stability would positively aid the machining process by providing a wider machining gap and better removal of debris, resulting in a more efficient and effective machining and flushing cycle [47]. In the present research, 1800 rpm is concluded as the optimum stirring velocity as facilitates minimum surface irregularities and maximum material removal.

4. Conclusions

In the present research, the impact of powder concentration, stirring velocity and pulse-off time during PMEDM of Ti-6Al-4V is experimentally investigated and the major findings are presented below:

- PC and stirring velocity are the primary determinants of the powder-mixed dielectric's stability, which in turn has a significant impact on EDM machining. With the optimal values for both, the machining cycle can be optimized to provide a high surface quality and MRR. It also contributes to enhanced conductivity which has improved the machining and flushing cycle due to the wider machining gap and greater debris removal caused by the enhanced conductivity.
- According to the ANOVA analysis, Pulse-off time individually and PC and stirring velocity in combination were found to have a significant effect on the MRR and SR during the PMEDM of Ti-6Al-4V.
- At the lower magnitude of powder concentration and stirring velocity (1 g/L and 900 rpm), an increase in pulse-off time from 28 μ s to 98 μ s has resulted in about 22 % improvement in surface finish. A longer pulse-off time ensures the machined region from overheating and keeps the machining conditions consistent by allowing more significant flushing and cooling.
- Increasing the powder concentration from 1 g/L to 2 g/L and stirring velocity from 900 rpm to 1800 rpm respectively has resulted in an improvement in the quality of SR by 24 % and MRR by 17 %. However, pushing the concentration further from 2 g/L to 3 g/L has a negative impact on the machining performance, it reduced SR by 14 % and MRR by 15 %. The outcomes of the experimental findings were further validated through grey relational analysis.

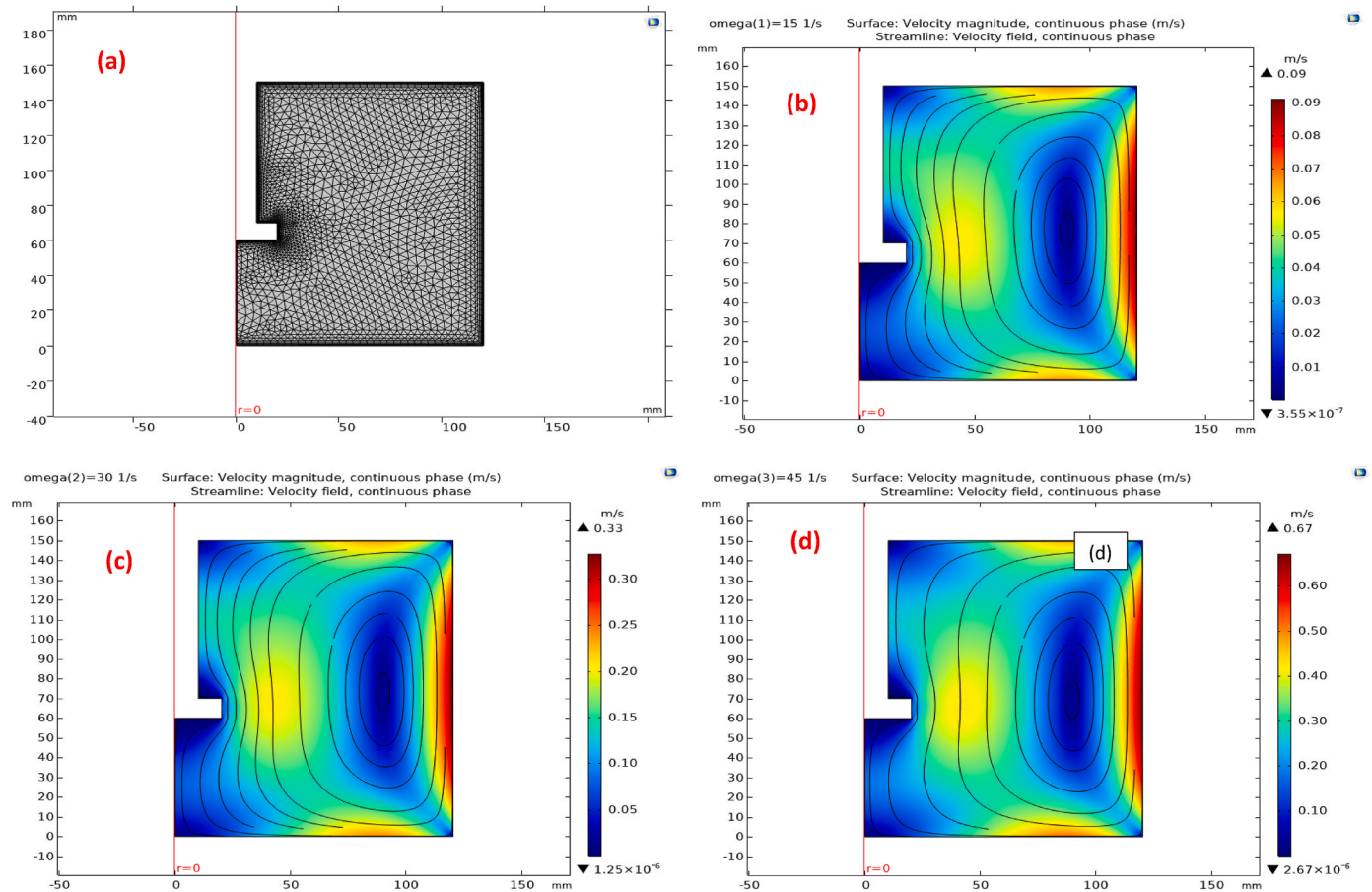


Fig. 12. (a) Dimension of stirrer and stirring tank with meshing, velocity distribution in the flow field (b) at 900 rpm (c) 1800 rpm (d) 2700 rpm.

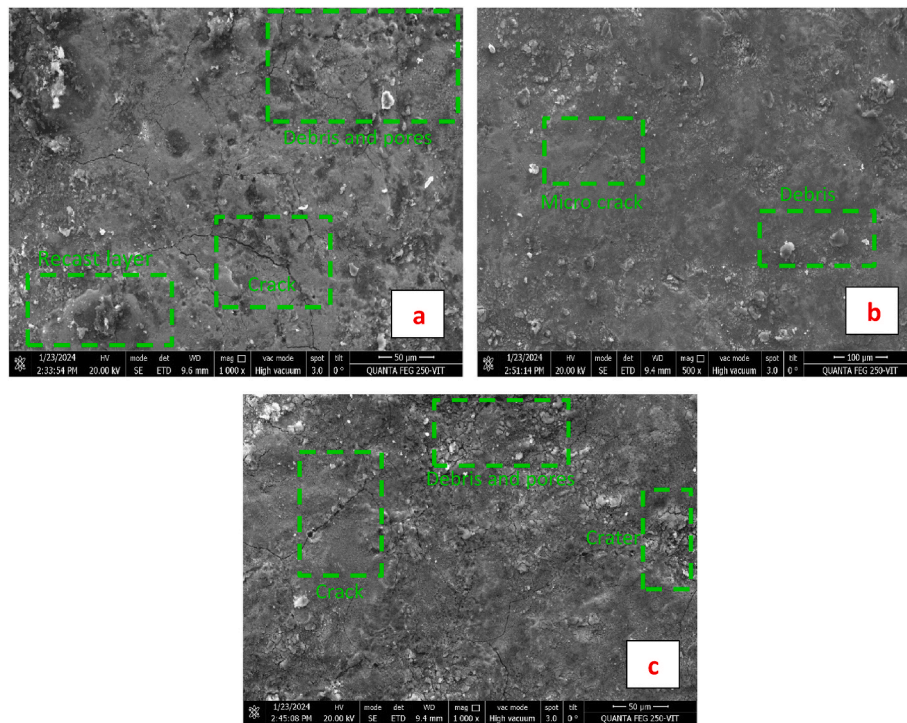


Fig. 13. (a–c) Surface morphology of machined surface at 2 g/L and a) 900 rpm b) 1800 rpm c) 2700 rpm stirring velocity with pulse off 28 μ s.

- The results of the scanning electron microscopy and three-dimensional surface analysis reflect the same pattern and trend that was seen in the experimental findings, and they additionally confirm that the experimental findings are valid.
- Lemon peel dielectric is suitable dielectric with CuO nanopowder with PC 2 g/L and 1800 rpm SV is the most suitable optimum combination to achieve a smooth surface with high MRR.
- From simulation analysis, it is observed that an increase in stirring velocity will increase fluid flow in the tank, which needs to be optimized to get the best results because excess stirring speed will cause Nano dielectric instability and compromise the machining cycle. Moreover, it is confirmed that all the stirring speeds have achieved the settling velocity of 6.913×10^{-9} m/s which was the minimum requirement to avoid the settling of particles in the tank.

CRedit authorship contribution statement

Rajnit Rakhshaskar: Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Kannan Chidambaram:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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